

September 2022 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains SEVEN questions. Answer ALL questions.

At the top of your answer scripts, please write the following:

- Your full name;
- Your class group; and
- Your class teacher's name.

Time Allowed: *2 hours*

You are supplied with: *No additional materials*

Calculators: *Calculators are NOT allowed in this examination*

1. **Total: 15 points (18 minutes)**

Suppose that a pair of fair dice is tossed. Let random variable X denote the sum of the two numbers obtained. That is, X can equal x , for $x = 2, 3, \dots, 12$.

- (a) Find $\Pr(X = x)$, for $x = 6$. **(3pts)**
- (b) Work out a general expression for $\Pr(X = x)$, for any x . (Alternatively, you may list the values of the probability function over its entire support.) **(4pts)**
- (c) Let A denote the event that “ X is divisible by 4”, and let B denote the event that “ X is greater than 9”. Find $\Pr(A)$, $\Pr(B)$, and $\Pr(A \cap B)$. **(4pts)**
- (d) State Bayes’ theorem in terms of events A and B . Verify numerically that it holds. **(4pts)**

Let D_{ij} be the event that die 1 shows the number i and that die 2 shows the number j , for $i, j = 1, \dots, 6$. Clearly, these 36 events form the finest partition of the set of possible outcomes. It follows that $\Pr(X = x) = \sum_{\{(i,j): i+j=x\}} \Pr(D_{ij})$, where $\Pr(D_{ij}) = 1/36$ for all i, j .

3 points. The answer for part (a) is:

$$\Pr(X = 6) = \Pr(D_{15}) + \Pr(D_{51}) + \Pr(D_{24}) + \Pr(D_{42}) + \Pr(D_{33}) = 5/36.$$

4 points. The answer for part (b) is:

$$\Pr(X = x) = \frac{\min\{(x-1), (13-x)\}}{36}$$

for $x = 2, 3, \dots, 12$.

4 points. The answer for part (c) is obtained by noting that:

$$A = (X = 4) \cup (X = 8) \cup (X = 12), \text{ and } B = (X = 10) \cup (X = 11) \cup (X = 12).$$

It follows that $\Pr(A) = 1/4$, $\Pr(B) = 1/6$, and $\Pr(A \cap B) = 1/36$.

4 points. The answer for part (d) is that:

Bayes’ theorem is given by

$$\Pr(A|B) = \Pr(B|A)\Pr(A)/\Pr(B).$$

To verify it numerically, we note that $\Pr(A|B) = \frac{1/36}{1/6} = 1/6$ and $\Pr(B|A) = \frac{1/36}{1/4} = 1/9$, so that $RHS = \frac{(1/9)(1/4)}{(1/6)} = 1/6 = LHS$.

2. Total: 15 points (18 minutes)

Consider the following density function,

$$f_{X,Y}(x,y) = \begin{cases} k(x^2 + y^2), & \text{for } 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What are the conditions under which $f_{X,Y}$ is a valid density function? **(5pts)**
- (b) Suppose you are interested in working out $\Pr(X < Y^2)$. Sketch the region (in the domain of $f_{X,Y}$) over which you would integrate $f_{X,Y}$ to find the solution. Identify axis labels, titles and label any lines/curves that you draw. **(5pts)**
- (c) Hence, or otherwise, show that $\Pr(X < Y^2) = 26/35$. **(5pts)**

5 points. The answer for part (a) is:

The density function must satisfy the conditions:

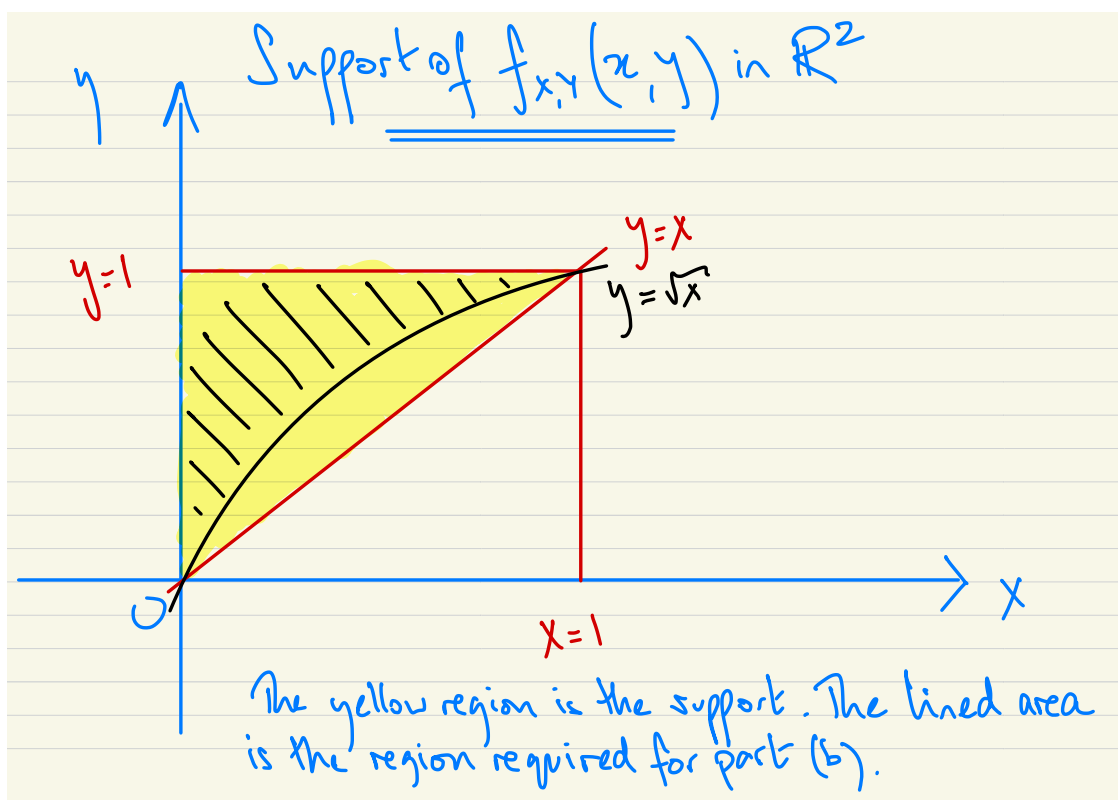
- $f_{X,Y}(x,y) \geq 0$ for all $(x,y) \in \mathbb{R}^2$;
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) \, dx \, dy = 1$.

The first condition is satisfied so long as $k \geq 0$. To work out the second condition, we have

$$\begin{aligned} k \int_0^1 \int_0^y (x^2 + y^2) \, dx \, dy &= k \int_0^1 \left[x^3/3 + xy^2 \right]_0^y dy \\ &= k \int_0^1 [y^3/3 + y^3] dy \\ &= k [y^4/12 + y^4/4]_0^1 \\ &= k/3. \end{aligned}$$

Then, setting $k/3 = 1$, we have that $k = 3$.

5 points. The answer for part (b) is:



5 points. The answer for part (c) is:

$$\begin{aligned}
 \Pr(X < Y^2) &= 3 \int_0^1 \int_{\sqrt{x}}^1 (x^2 + y^2) dy \, dx \\
 &= 3 \int_0^1 \left[x^2 + 1/3 - x^{5/2} - x^{3/2}/3 \right] dx \\
 &= 3[1/3 + 1/3 - 2/7 - 2/15] = 26/35.
 \end{aligned}$$

3. Total: 10 points (12 minutes)

Consider random variable X with probability function

$$\Pr(X = x) = \begin{cases} (1 - \pi)^x \pi, & \text{for } x = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Show that the moment-generating function of X , say $M_X(t)$, is given by

$$M_X(t) = \frac{\pi}{1 - e^t(1 - \pi)},$$

and state values of t for which $M_X(t)$ is well-defined.

(5pts)

(b) Use $M_X(t)$ to derive $E(X)$.

(5pts)

5 points. The answer for part (a) is:

$$\begin{aligned} M_X(t) &= E(e^{tX}) = \sum_{x=0}^{\infty} e^{tx} \Pr(X = x) \\ &= \sum_{x=0}^{\infty} e^{tx} (1 - \pi)^x \pi \\ &= \pi \sum_{x=0}^{\infty} (e^t (1 - \pi))^x \\ &= \frac{\pi}{1 - e^t(1 - \pi)}, \end{aligned}$$

for $t < -\log(1 - \pi)$.

5 points. The answer for part (b) is derived as follows:

$$\begin{aligned} M'_X(t) &= \frac{\pi(1 - \pi)e^t}{(1 - e^t(1 - \pi))^2}, \text{ so} \\ M'_X(0) &= \frac{1 - \pi}{\pi} = E(X). \end{aligned}$$

4. **Total: 15 points (18 minutes)**

Consider an estimator $\hat{\theta}_N$ for an unknown parameter θ , where N denotes the sample size.

- (a) Define formally the properties of bias, variance and mean-square error of $\hat{\theta}_N$.
Derive an exact relationship between these three properties. **(5pts)**
- (b) Define formally what is meant by consistency of $\hat{\theta}_N$ (**without** using the “ p lim” operator). **(4pts)**
- (c) Suppose we have a random sample of size $N \geq 3$ given by $\{X_i\}_{i=1}^N$ from the $\text{Normal}(\mu, 1)$ distribution. Let $\theta = \mu^2$, and let $\bar{X}_N$ denote the sample mean. Your colleague suggests:

$$\begin{aligned}\hat{\mu} &= X_1 + (X_2 - X_3)^2 - 2, \text{ as an estimator for } \mu; \text{ and} \\ \hat{\theta}_N &= \bar{X}_N^2 - 1/N, \text{ as an estimator for } \theta.\end{aligned}$$

Assess whether each estimator, $\hat{\mu}$ and $\hat{\theta}_N$, is unbiased and consistent. **(6pts)**

5 points. The answer to part (a) is:

$$\begin{aligned}\text{Bias}(\hat{\theta}_N) &= \text{E}(\hat{\theta}_N) - \theta \\ \text{Var}(\hat{\theta}_N) &= \text{E}\left(\left(\hat{\theta}_N - \text{E}(\hat{\theta}_N)\right)^2\right) \\ \text{MSE}(\hat{\theta}_N) &= \text{E}\left(\left(\hat{\theta}_N - \theta\right)^2\right)\end{aligned}$$

so that, using the binomial theorem and linearity of expectations, we have

$$\begin{aligned}\text{MSE}(\hat{\theta}_N) &= \text{E}\left(\left(\left(\hat{\theta}_N - \text{E}(\hat{\theta}_N)\right) + \left(\text{E}(\hat{\theta}_N) - \theta\right)\right)^2\right) \\ &= \text{Var}(\hat{\theta}_N) + \text{Bias}^2(\hat{\theta}_N) + 2\text{E}\left(\left(\hat{\theta}_N - \text{E}(\hat{\theta}_N)\right)\left(\text{E}(\hat{\theta}_N) - \theta\right)\right) \\ &= \text{Var}(\hat{\theta}_N) + \text{Bias}^2(\hat{\theta}_N).\end{aligned}$$

4 points. For part (b), we say that $\hat{\theta}_N$ is a consistent estimator for θ if for any (arbitrarily small) $\epsilon > 0$, it holds that

$$\lim_{N \rightarrow \infty} \Pr\left(|\hat{\theta}_N - \theta| > \epsilon\right) = 0.$$

6 points. The answers to part (c) are:

$$\begin{aligned} E(\hat{\mu}) &= \mu - 2 + E(X_2^2 - 2X_2X_3 + X_3^2) \\ &= \mu - 2 + 2(1 + \mu^2) - 2\mu^2 = \mu \end{aligned}$$

since $E(X_2X_3) = E(X_2)E(X_3)$ under independence. Hence, $\hat{\mu}$ is an unbiased estimator for μ . It cannot be consistent since its value does not depend on N .

Next,

$$\begin{aligned} E(\hat{\theta}_N) &= E(\bar{X}^2 - 1/N) \\ &= \mu^2 + 1/N - 1/N = \mu^2 \end{aligned}$$

since $\text{Var}(\bar{X}) = 1/N$. We know by Chebyshev's LLN that $\bar{X} \xrightarrow{p} \mu$ and by the continuous mapping theorem that $\bar{X}^2 \xrightarrow{p} \mu^2$. Moreover, since $\lim_{N \rightarrow \infty} 1/N = 0$, it follows that $\hat{\theta}_N$ is a consistent estimator for θ .

5. **Total: 20 points (24 minutes)**

- (a) Suppose we have a set of mutually independent random variables $X_i \sim \text{Normal}(0, \sigma^2)$ for $i = 1, \dots, N$, where σ^2 is unknown.
- i. Derive the method of moments (MM) estimator for σ^2 . **(2pts)**
 - ii. Derive the maximum likelihood (ML) estimator for σ^2 . (You do **not** need to state the second-order condition.) **(6pts)**
- (b) Now suppose we have a different set of mutually independent random variables $Y_i \sim \text{Normal}(i\mu, i^3\sigma^2)$ for $i = 1, \dots, N$, where μ is unknown but σ^2 is known.
- i. Derive the ML estimator for μ . (You do **not** need to state the second-order condition.) **(10pts)**
 - ii. Comment on the distribution of the ML estimator. **(2pts)**

Hint: You may find it helpful to recall that if $X \sim \text{Normal}(\mu, \sigma^2)$, then

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \text{ for any } x \in \mathbb{R}.$$

2 points. For part (a)i.:

We define $\hat{\sigma}_{MM}^2$ such that $N^{-1} \sum_{i=1}^N X_i^2 = \hat{\sigma}_{MM}^2 + (\mathbb{E}(X_i))^2$ so that $\hat{\sigma}_{MM}^2 = N^{-1} \sum_{i=1}^N X_i^2$.

6 points. For part (a)ii.:

$$\mathcal{L}(\sigma^2; \mathbf{x}) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x_i^2}{2\sigma^2}} \propto (\sigma^2)^{-N/2} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^N x_i^2}$$

$$\ell(\sigma^2; \mathbf{x}) = -\frac{N}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^N x_i^2 + \text{constants w.r.t } \sigma^2$$

$$s(\sigma^2; \mathbf{x}) = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^N x_i^2$$

Setting $s(\hat{\sigma}_{ML}^2; \mathbf{X}) = 0$ gives $\hat{\sigma}_{ML}^2 = \frac{1}{N} \sum_{i=1}^N X_i^2$, which is the ML estimator.

10 points. For part (b)i.:

$$\begin{aligned} \mathcal{L} &= \prod_{i=1}^N \frac{1}{\sqrt{2\pi(i^3\sigma^2)}} \exp\left(-\frac{(y_i - i\mu)^2}{2i^3\sigma^2}\right) \\ &= (2\pi)^{-N/2} \sigma^{-N} \left(\prod_{i=1}^N i^{-3/2}\right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^N i^{-3}(y_i - i\mu)^2\right), \text{ so} \end{aligned}$$

$$\log \mathcal{L} = -\frac{N}{2} \log 2\pi - N \log \sigma - \frac{3}{2} \sum_{i=1}^N \log i - \frac{1}{2\sigma^2} \sum_{i=1}^N i^{-3}(y_i - i\mu)^2, \text{ thus,}$$

$$\frac{\partial \log \mathcal{L}}{\partial \mu} = -\frac{1}{\sigma^2} \sum_{i=1}^N i^{-3}(y_i - i\mu)(-i).$$

Setting the score to zero implies that

$$\begin{aligned} \hat{\mu}_{ML} \sum_{i=1}^N i^{-1} &= \sum_{i=1}^N i^{-2} Y_i, \text{ so that our ML estimator is} \\ \hat{\mu}_{ML} &= \frac{\sum_{i=1}^N i^{-2} Y_i}{\sum_{i=1}^N i^{-1}}. \end{aligned}$$

2 points. For part (b)ii.:

Since $\hat{\mu}_{ML}$ is a linear combination of mutually independent normal random variables, it is itself normally distributed.

6. **Total: 10 points (12 minutes)**

You have several identical-looking coins in your pocket; most are fair coins, but there is one coin that comes up heads with probability $1/4$. You pick a coin at random and flip it 10 times. You decide that you will reject the hypothesis that the coin is fair if it comes up heads 3 times or fewer.

- (a) Set this up as a formal hypothesis test. What is the null hypothesis, alternative hypothesis, test statistic, and decision rule? **(4pts)**
- (b) Derive explicit algebraic expressions for the size and power of this test (but do **not** attempt to solve for their exact numerical values). **(4pts)**
- (c) Suppose that the coin comes up heads only once. Derive an explicit algebraic expression for the corresponding p -value (but do **not** attempt to solve for its exact numerical value). **(2pts)**

Hint: You may find it helpful to recall that if $X \sim \text{Binomial}(n, p)$, then

$$\Pr(X = x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \dots, n, \\ 0, & \text{otherwise.} \end{cases}$$

4 points. For part (a):

We observe the random sample $Y_1, \dots, Y_{10} \sim \text{Bernoulli}(p)$, where $Y_i = 1$ if the coin comes up heads on the i^{th} toss. We want to test $H_0 : p = 1/2$ versus $H_1 : p = 1/4$. The test statistic is the total number of heads, $X = \sum_{i=1}^{10} Y_i \sim \text{Bernoulli}(10, p)$. The decision rule is to reject H_0 when $X \leq 3$.

4 points. For part (b):

We are testing a simple null hypothesis versus a simple alternative, so it is straightforward to compute the size and power. The size is the probability of a Type I error, that is,

$$\begin{aligned} \alpha &= \Pr_{H_0}(X \leq 3) = \sum_{j=0}^3 \binom{10}{j} \left(\frac{1}{2}\right)^j \left(1 - \frac{1}{2}\right)^{10-j} \\ &= 0.001 + 0.010 + 0.044 + 0.117 = 0.172. \end{aligned}$$

Students are NOT expected to provide the numbers in the previous line.

The power is 1 minus the probability of a Type II error,

$$\begin{aligned}\beta &= \Pr_{H_1}(X \leq 3) = \sum_{j=0}^3 \binom{10}{j} \left(\frac{1}{4}\right)^j \left(1 - \frac{1}{4}\right)^{10-j} \\ &= 0.056 + 0.188 + 0.282 + 0.250 = 0.776.\end{aligned}$$

Students are NOT expected to provide the numbers in the previous line.

2 points. For part (c):

The p -value is the probability of observing a result *at least as extreme* as $X = 1$ (assuming H_0 is true), in other words, the probability of the coin coming up heads one *or zero* times,

$$\begin{aligned}\alpha &= \Pr_{H_0}(X \leq 1) = \sum_{j=0}^1 \binom{10}{j} \left(\frac{1}{2}\right)^j \left(1 - \frac{1}{2}\right)^{10-j} \\ &= 0.001 + 0.010 = 0.011.\end{aligned}$$

Students are NOT expected to provide the numbers in the previous line.

7. Total: 15 points (18 minutes)

- (a) Explain *carefully* what is meant by a $100(1 - \alpha)\%$ confidence interval (CI) for an unknown parameter μ , for some $\alpha \in (0, 1)$. **(2pts)**
- (b) Suppose we have a random sample $\{X_i\}_{i=1}^N$ from a distribution with finite mean μ and finite positive variance σ^2 . Provide a formal statement of the (Lindeberg-Lévy) central limit theorem (CLT). **(2pts)**
- (c) Suppose we have a random sample $\{X_i\}_{i=1}^N$ from a $\text{Normal}(\mu, \sigma^2)$ distribution with finite mean μ and finite positive variance σ^2 , where σ^2 is known.
 - i. Derive a $100(1 - \alpha)\%$ CI for μ **without** relying on the CLT. **(5pts)**
 - ii. How could one use the CLT to conduct inference on μ ? How does the CLT-based approach compare with the method in part (i) above? **(3pts)**
 - iii. Now assume σ^2 is unknown. Explain how to conduct inference on μ in this case. Your answer should explain whether the CLT is needed or not. **(3pts)**

2 points. For part (a):

A $100(1 - \alpha)\%$ CI refers to an interval estimator which, in hypothetical repeated samples, covers the true value of μ in $100(1 - \alpha)\%$ of cases. [No marks to be awarded for answers which say things such as “*probability that μ falls within the CI is...*”. In other words, no marks for answers that imply that μ is random.]

2 points. For part (b):

Under the conditions outlined in the question, it holds that $\sqrt{N}(\bar{X}_N - \mu) \xrightarrow{d} \text{Normal}(0, \sigma^2)$ as $N \rightarrow \infty$.

5 points. For part (c)i.:

Given the pivotal function $\sqrt{N}(\bar{X}_N - \mu)/\sigma \sim \text{Normal}(0, 1)$ for any N , we can find always find a q_1 and q_2 (quantiles on the CDF of the standard normal distribution) such that

$$\Pr\left(q_1 < \sqrt{N}(\bar{X}_N - \mu)/\sigma < q_2\right) = 1 - \alpha$$

for some $\alpha \in (0, 1)$. Suppose $q_1 = z_{\alpha/2}$ and $q_2 = z_{1-\alpha/2}$ to denote specific values obtained from statistical tables. Given symmetry of the standard normal distribution, we know further that $z_{\alpha/2} = -z_{1-\alpha/2}$. Rearranging the terms in the above inequalities and plugging in for q_1 and q_2 , we have

$$\Pr\left(\bar{X}_N - z_{1-\alpha/2} \frac{\sigma}{\sqrt{N}} < \mu < \bar{X}_N + z_{1-\alpha/2} \frac{\sigma}{\sqrt{N}}\right) = 1 - \alpha$$

The above result is motivation for us to construct an interval estimator given by

$$\left[\bar{X}_N - z_{1-\alpha/2} \frac{\sigma}{\sqrt{N}}, \bar{X}_N + z_{1-\alpha/2} \frac{\sigma}{\sqrt{N}}\right]$$

which we refer to as the $100(1 - \alpha)\%$ CI for μ .

3 points. For part (c)ii.:

We would use the *convergence in d* result from part (b) – for instance,

$$\sqrt{N}(\bar{X}_N - \mu)/\sigma \xrightarrow{d} \text{Normal}(0, 1)$$

as $N \rightarrow \infty$ to identify a pivotal function for an asymptotically valid $100(1 - \alpha)\%$ CI. The remaining steps are identical to part (c)i.

The availability of such an approach is invaluable for inference in situations where the distributional family is unknown. Nevertheless, a CLT-based approach is redundant in our current setting (and should not be used) because we have (as explained in part (c)i.) an exact distributional result that is valid for any N .

3 points. For part (c)iii.:

The CLT-based approach should still not be used. Indeed, while the approach of part (c)i. can no longer be used (since σ^2 is unknown), we have another exact distributional result available to us. Indeed, we know that

$$\sqrt{N}(\bar{X}_N - \mu)/\hat{\sigma}_u \sim t_{N-1},$$

for any N , where $\hat{\sigma}_u = (N-1)^{-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2$. This yields a pivotal function to be used in the same manner as explained in part(c)i. to construct a CI for μ .